

M001 Week 2 — English Worksheet

Topic: Loops · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
repeat 4 times:
  place block down
  move forward
```

What does the agent build? How many blocks?


```
repeat 6 times:
  place block down
  move up by 1
```

Is this a wall lying flat, or a tower standing up? How tall?


```
repeat 5 times:
  move forward
```

Does the agent place any blocks? Where does it end up?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to use a loop to place a row of **5 blocks**.

```
place block down
move forward
place block down
move forward
place block down
move forward
place block down
move forward
place block down
```

Write the fixed code using a loop:

Why was the original code messy? Why does your loop work?

Bug B — The agent is supposed to build a tower **5 blocks tall**, with one block at each height.

```
repeat 5 times:
  place block down
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to build a row of 6 blocks, all in a straight line on the ground.

```
repeat 6 times:  
  place block down  
  move forward  
  turn right
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Use a loop to build a **20-block straight wall**. Then build a **U-shaped wall** (three sides). When you finish, come back here.

Send a photo or video of what you built, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My loop repeated ... times.

Inside the loop, the agent ...

The hardest part was ...

To fix it, I ...

A loop is useful because ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while you walk the camera around what you built. Talk through it like you are teaching someone who has never seen it. Try to use these words: **loop, repeat, times, block, agent.**

1. Show what your loop built.
2. Read your loop out loud and say what it does.
3. Show one bug you hit and how you fixed it.
4. Say why a loop is better than writing the same lines over and over.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.